

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
aaharlamova@aaharlamova:~$ ls /etc/ > file.txt
aaharlamova@aaharlamova:~$ ls >> file.txt
aaharlamova@aaharlamova:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
```

Рисунок 1: Запись в файл

```
aaharlamova@aaharlamova:~$ grep .conf file.txt > conf.txt
aaharlamova@aaharlamova:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
```

Рисунок 2: Поиск расширения .conf

```
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e03/report/_resources/csl
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e03/report/bib/cite.bib
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e04/report/_resources/csl
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e04/report/bib/cite.bib
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e05/report/_resources/csl
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e05/report/bib/cite.bib
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e06/report/_resources/csl
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stag
e06/report/bib/cite.bib
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_
resources/csl
/home/aaharlamova/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/b
ib/cite.bib
/home/aaharlamova/git-extended/.git/hooks/commit-msg.sample
/home/aaharlamova/git-extended/.git/objects/ba/c1889e6720954f7e40f48cb7c417e00653b8e8
/home/aaharlamova/git-extended/.git/config
/home/aaharlamova/conf.txt
aaharlamova@aaharlamova:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpcups.conf
```

Рисунок 4: Поиск файлов

```
aaharlamova@aaharlamova:~$  
aaharlamova@aaharlamova:~$ find /etc -name "h*" -print | less  
aaharlamova@aaharlamova:~$ find ~ -name "log*" > logfile &  
[1] 18740  
aaharlamova@aaharlamova:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
aaharlamova@aaharlamova:~$ rm logfile  
aaharlamova@aaharlamova:~$
```

Рисунок 5: Фоновый запуск процесса

```
aaharlamova@aaharlamova:~$  
aaharlamova@aaharlamova:~$ gedit &  
[1] 19029  
aaharlamova@aaharlamova:~$ ps | grep gedit  
19029 pts/1    00:00:00 gedit  
aaharlamova@aaharlamova:~$ kill 19029  
[1]+  Terminated                  gedit  
aaharlamova@aaharlamova:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    --direct
        Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually
        smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indi-
        rect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576
        bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
aaharlamova@aaharlamova:~$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 41758800 106612112 29% /
devtmpfs         4011296          0   4011296  0% /dev
tmpfs            4035672         92   4035580  1% /dev/shm
tmpfs            1614272        1960   1612312  1% /run
tmpfs             1024           0        1024  0% /run/credentials/systemd-journald.service
/dev/nvme0n1p3 155186176 41758800 106612112 29% /home
tmpfs            4035672       132916   3902756  4% /tmp
/dev/nvme0n1p2 1992552    518800   1352512 28% /boot
/dev/loop0        68480       68480          0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1       115712     115712          0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop2        49280      49280          0 100% /var/lib/snapd/snap/snapd/25935
tmpfs             1024           0        1024  0% /run/credentials/systemd-resolved.service
tmpfs            807132        188   806944  1% /run/user/1094
aaharlamova@aaharlamova:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/77
4      ./git-extended/.git/objects/ba
4      ./git-extended/.git/objects/58
4      ./git-extended/.git/objects/d8
4      ./git-extended/.git/objects/e5
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
971224 .
aaharlamova@aaharlamova:~$
```

Рисунок 10: Запуск команды du

```
/home/aaharlamova/git-extended/.git/objects/db
/home/aaharlamova/git-extended/.git/objects/77
/home/aaharlamova/git-extended/.git/objects/ba
/home/aaharlamova/git-extended/.git/objects/58
/home/aaharlamova/git-extended/.git/objects/d8
/home/aaharlamova/git-extended/.git/objects/e5
/home/aaharlamova/git-extended/.git/logs
/home/aaharlamova/git-extended/.git/logs/refs
/home/aaharlamova/git-extended/.git/logs/refs/heads
/home/aaharlamova/git-extended/.git/logs/refs/remotes
/home/aaharlamova/git-extended/.git/logs/refs/remotes/origin
/home/aaharlamova/monthly
/home/aaharlamova/reports
/home/aaharlamova/reports/monthly
/home/aaharlamova/reports/monthly/monthly
/home/aaharlamova/ski.plases
/home/aaharlamova/ski.plases/equipment
/home/aaharlamova/ski.plases/plans
/home/aaharlamova/australia
/home/aaharlamova/play
/home/aaharlamova/play/games
/home/aaharlamova/play/games/play
aaharlamova@aaharlamova:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.